

# App Dev Project Report

## 1. Student Details

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**About Me:** I am a Student at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

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## 2. Project Details

**Project Title:** Trekking Management Application

### **Problem Statement:**

The Trekking Management Application is a simple web app that helps manage trekking activities in one place. It allows admins to manage treks, staff to handle operations, and users to book and track their treks. The system makes the booking process easier, avoids overbooking, and replaces manual record keeping.

### **Approach:**

I started by understanding the project requirements and the features needed for each user role. After that, I designed the database to store user, trek, and booking information. Then, I developed the application using Flask for the backend and HTML, CSS, Bootstrap, and Jinja2 for the frontend. Finally, I tested the application to make sure all the features were working correctly.

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## 3. AI/LLM Declaration

I used ChatGPT (GPT-5) for the initial Flask project setup, database design suggestions, debugging, code improvements, Bootstrap class suggestions, and writing parts of the project report. The overall AI/LLM usage in this project was approximately **15-20%**. All major implementation, testing, and final integration were completed by me.

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## 4. Technologies and Frameworks Used

| Technology / Library | Purpose                                                  |
|----------------------|----------------------------------------------------------|
| Flask                | Core backend web framework                               |
| SQLAlchemy           | Object Relational Mapper for SQLite database             |
| Jinja2               | Template engine for rendering dynamic HTML pages         |
| Bootstrap 5          | Frontend styling and responsive design                   |
| Werkzeug Security    | Password hashing and verification for secure login       |
| Flask-Login          | User authentication and session management               |
| SQLite               | Lightweight local database for storing user data         |
| python-dotenv        | Loads environment variables from a <code>env</code> file |

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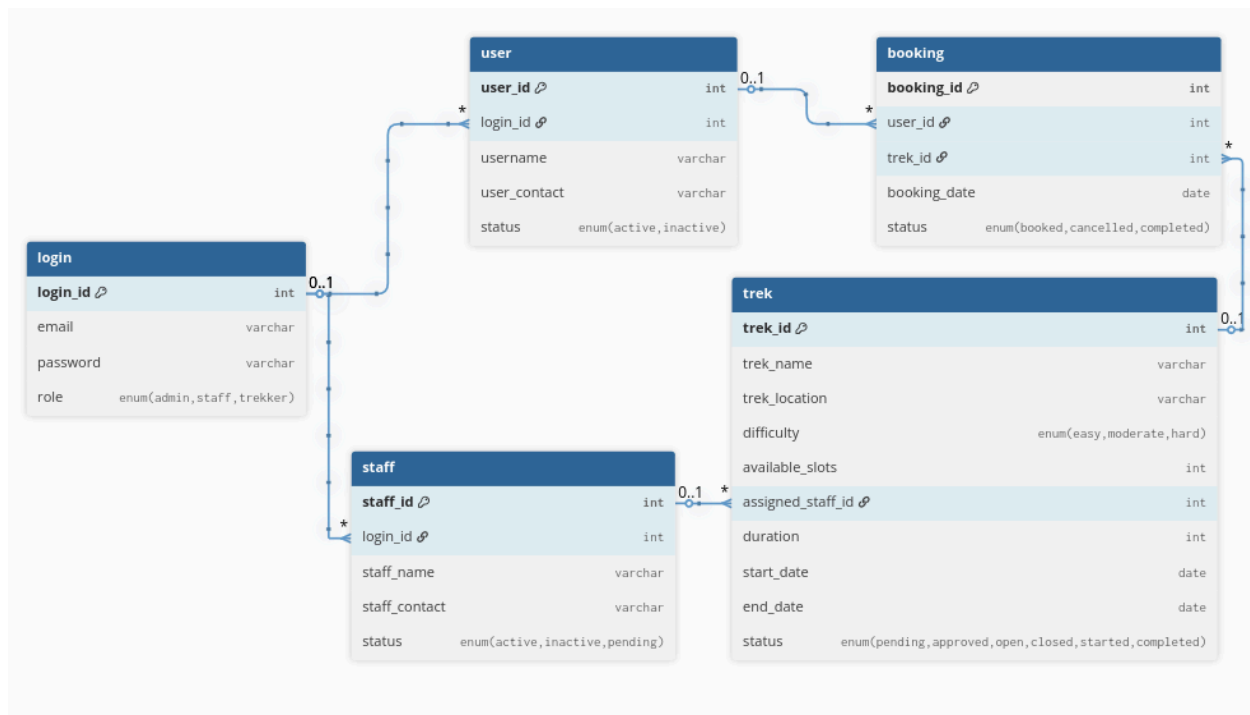
## 5. Database Schema / ER Diagram

### Tables:

1. **Login** – Stores login details (login\_id, email, password, role)
2. **User** – Stores user account details (id, login\_id, name, contact, status)
3. **Staff** – Stores staff information (id, login\_id, name, contact, status)
4. **Trek** – Stores trek information (id, name, location, difficulty, available\_slots, assigned\_staff\_id, duration, start\_date, end\_date, status)
5. **Booking** – Stores trek booking details (id, user\_id, trek\_id, booking\_date, status)

### Relationships:

- One-to-One → User → Login
- One-to-One → Staff → Login
- One-to-Many → User → Booking
- One-to-Many → Trek → Booking
- One-to-Many → Trek → Staff



## 7. Architecture and Features

### Architecture Overview:

- `app.py` – Main entry point of the Flask application.
- `app/__init__.py` – Initializes the Flask app, extensions, and registers Blueprints.
- `config.py` – Stores application configuration such as database URI and secret key.
- `models/` – Contains SQLAlchemy database models.
- `routes/` – Contains Flask Blueprints for Admin, User, Staff, and Authentication routes.
- `templates/` – Contains Jinja2 HTML templates for the user interface.
- `static/` – Contains CSS, JavaScript, and other static assets.

### Implemented Features:

- User and Staff registration with secure login authentication
- Role-based dashboards for Admin, Staff, and Trekker
- Trek creation, management, and staff assignment by Admin
- Trek booking with overbooking prevention
- Staff approval and user/staff blacklist management
- Trek search and filtering by location and difficulty
- Booking history and trek status tracking

- Responsive user interface built using Bootstrap
  - Secure password hashing and session management using Flask-Login
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## 8. Video Presentation

**Drive Link:**

 23f3002456\_video.mp4